

Alvaro Perdomo Gutierrez

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across fintech, e-commerce, and healthcare, building scalable Python backend systems and REST APIs with strong cloud infrastructure expertise. Experienced in API design, database optimization, microservices deployment, and cross-functional collaboration on international distributed teams using FastAPI, PostgreSQL, and AWS.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS
Backend: FastAPI, Django, Flask, Node.js, Express, RabbitMQ, Kafka
Database: PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch, TcaplusDB
Testing: Pytest, Jest, Cypress, Playwright, k6
Cloud & DevOps: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, GitHub Actions, Terraform, Linux, CloudWatch
Others: RESTful API, GraphQL, Git, GitHub, Microservices, Serverless Architecture, CI/CD, Infrastructure as Code, Agile, Scrum, Jira

EXPERIENCE

Leanware May 2024–Present
Senior Software Engineer

- Led the development of the 2Moon Capital algorithmic trading platform backend using **Python** and **FastAPI**, handling real-time market data ingestion and position management across distributed microservices that processed over **100k+ trade events daily**.
- Designed and implemented RESTful APIs for the Reachr email outreach automation system, integrating **PostgreSQL** for campaign tracking and **Redis** for rate-limiting and session caching, enabling the product team to onboard **50+ enterprise customers** within the first quarter.
- Architected the backend infrastructure for the PRD Agent AI-powered product document generator, deploying containerized **Python** services on **AWS ECS** with **Kubernetes** orchestration, reducing API response latency from **2.4s to 680ms** through query optimization and connection pooling.
- Collaborated with the frontend team to design API contracts and schema validation for the CodiQ GitHub analytics platform, using **TypeScript** type definitions to ensure consistency and reducing integration bugs by **32%** across quarterly releases.

- Optimized database queries across multiple **PostgreSQL** schemas used by the financial reporting system, adding strategic indexes and query batching that cut dashboard load times from **5.2s to 1.1s** for users analyzing trading performance.
- Implemented automated testing suites using **Pytest** for the trading and outreach platforms, achieving **78% code coverage** and catching edge-case failures in market data parsing before production deployment.
- Maintained and enhanced CI/CD pipelines using **GitHub Actions** for all backend services, parallelizing test execution and containerization steps to reduce deployment cycles from **18 minutes to 6 minutes**, improving team velocity on rapid feature iterations.
- Participated in on-call rotation and incident response, troubleshooting production issues in market data feeds and API services, implementing structured logging with **CloudWatch** to reduce mean time to resolution by **40%** on recurrent failures.
- Mentored two junior backend engineers on **Python** best practices, API design patterns, and **PostgreSQL** optimization techniques, conducting weekly code reviews and pairing on complex distributed system challenges.
- Contributed to infrastructure-as-code migration using **Terraform**, codifying **AWS** resource provisioning for development, staging, and production environments across the trading and analytics platforms, reducing manual infrastructure overhead by **60%**.
- Designed the data ingestion pipeline for the financial platform using **Kafka** topics to decouple market data producers from consumer services, enabling independent scaling and reducing coupling between trading and reporting components.
- Owned the authentication and authorization system for multiple internal tools, implementing OAuth2 token validation in **FastAPI** middleware and securing API endpoints with role-based access control matching Leanware's team structure.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the backend architecture for the AI Shopping image search feature by building **Python** and **FastAPI** microservices that processed image embeddings from the proprietary visual recognition system, enabling users to search inventory by photo and driving **15% improvement in sell-through rates**.
- Integrated the Databricks platform migration by designing new **PostgreSQL** schemas for ML model metadata and training artifacts, reducing model deployment time from **2 quarters to 1 month** and cutting MVP iteration cycles from **2 weeks to 2 days** for the ML team.
- Built RESTful APIs exposing the resale-pricing algorithm and garment auto-photography processing to the mobile and web clients, using **FastAPI** request validation and **Redis** caching to handle **100k+ daily image processing requests** without exceeding latency budgets.
- Optimized the database layer by refactoring legacy **PostgreSQL** queries and adding materialized views for personalization ML features, reducing query times for single-SKU recommendations from **800ms to 240ms** across millions of daily requests.

- Implemented comprehensive logging and monitoring using **CloudWatch** and custom **Python** structured logging, enabling the team to detect a seller processing delay issue that was impacting **2023 complaints** and resolve it within **48 hours**.
- Collaborated with data scientists to deploy the garment measurement ML model into production using **Docker** containers and **Kubernetes** pods on **AWS ECS**, managing resource allocation and auto-scaling to handle variable traffic patterns.
- Developed and maintained internal backend tooling using **Python** and **Flask** for data validation and export workflows, reducing data QA cycle time by **30%** and improving team confidence in catalog accuracy.
- Conducted technical interviews and hiring for the backend engineering team, helping recruit five senior engineers and establishing consistent evaluation criteria aligned with the platform's complexity and scale requirements.
- Established testing standards using **Pytest** and integration test suites, enforcing **80% code coverage** requirements and reducing production bugs related to API contract violations by **45%** quarter over quarter.
- Participated in architecture discussions around microservices decomposition, advocating for **GraphQL** federation layer to reduce client payload sizes and improving API flexibility for the rapidly evolving mobile team.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in scaling the precision-matching ML platform backend using **Python** and **Django** to support **6x growth** from **~2 million members** while preserving clinical validity and HIPAA compliance across all algorithms and data flows.
- Designed and deployed the global Care Navigation platform backend using **FastAPI**, building RESTful APIs that powered member matching to mental health providers with configurable clinical criteria, supporting **Fortune 500 customers** including PepsiCo, Instacart, and Bain.
- Engineered the family launch feature released in September 2021, implementing jurisdiction-aware matching logic, dependent eligibility validation, and localization support using **PostgreSQL** stored procedures and **Redis** caching to handle real-time eligibility checks across **50+ US states**.
- Scaled HIPAA-compliant infrastructure by architecting secure **AWS** deployment patterns with encryption at rest and in transit, enabling onboarding of enterprise customers with strict compliance requirements and reducing security audit cycle time by **35%**.
- Built operational ML pipelines for dropout-risk prediction and demand forecasting using **Python** data processing frameworks, enabling the business team to proactively intervene with at-risk members and improve retention metrics.
- Optimized **PostgreSQL** queries and schema design for the member and provider databases, adding indexes and query batching that improved Care Navigation API response times from **1.2s to 380ms** and reduced database CPU utilization by **40%**.
- Implemented comprehensive API documentation using **FastAPI** OpenAPI schema generation and maintained Postman collections for partner integrations, improving onboarding time for integration partners by **50%** and reducing support tickets related to API confusion.

- Established CI/CD practices using **GitHub Actions** for automated testing and deployment, enabling daily releases to staging environments and weekly production deployments with **zero-downtime** deployment strategies.
- Contributed to incident response protocols and on-call rotations, implementing structured logging and monitoring with **CloudWatch** to ensure member-facing APIs maintained **99.9% uptime** during product scaling.
- Mentored junior engineers on backend architecture patterns and **Django** best practices, conducting code reviews and pairing on complex matching algorithm implementations in clinical care workflows.

Tencent Games

Nov 2014–Nov 2020

Software Engineer

- Built backend services for the Honor of Kings game platform using **Python** and **C++** that scaled to support **100 million daily active users** by November 2020, handling real-time matchmaking, leaderboards, and social features across distributed **Google Cloud Platform** infrastructure.
- Developed core gameplay APIs using **Python** and **FastAPI** for PUBG Mobile, processing **100+ players per match** with real-time state synchronization and enabling cross-platform play between mobile and console clients, supporting **tens of millions of concurrent connections**.
- Designed and optimized the TcaplusDB integration layer for persistent storage, managing **40 million+ daily player data records** and reducing database query latencies from **200ms to 45ms** through schema optimization and connection pooling strategies.
- Implemented anti-cheat infrastructure contributions using kernel-level detection logic to prevent unauthorized access and cheating behavior, working with the ACE team to maintain competitive integrity across millions of players and reducing reported abuse incidents by **60%**.
- Built the Tencent Gaming Buddy emulator backend in **Python**, providing lightweight game emulation for testing and allowing players on lower-end devices to experience games optimized for high-performance platforms.
- Collaborated across teams to globalize the Arena of Valor platform, adapting region-specific payment systems, localization logic, and compliance requirements into the core **PostgreSQL** and **Redis** data layers supporting players in **100+ countries**.
- Implemented comprehensive monitoring and alerting using custom **Python** telemetry collectors and Grafana dashboards, enabling rapid detection of gameplay issues and reducing mean time to incident resolution from **45 minutes to 10 minutes** for player-facing outages.
- Participated in major architecture migrations, helping transition legacy monolithic services to microservices using **Docker** containers and orchestration patterns, improving deployment frequency from **monthly to daily** and enabling faster feature iteration.
- Established testing frameworks and automated integration test suites for backend services, catching regressions in gameplay logic and preventing **95%** of critical bugs from reaching production environments.
- Mentored junior engineers on game server architecture, real-time systems design, and **Python** optimization techniques, contributing to the Tencent Institute of Games R&D knowledge base and training program.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014